Supermarket data, turned into decisions

Automating the path from raw CSV exports to sales insight and inventory action.

RAW DATA

BUSINESS ACTION

Made with GAMMA

Manual preparation takes time before analysis begins

01 Manual cleanup slows the start

Staff may clean inconsistent text, find missing values, remove duplicates, and prepare recurring reports manually.

02 Inconsistent handling weakens decisions

Manual steps may lead to inconsistent results.

03 Make the workflow repeatable

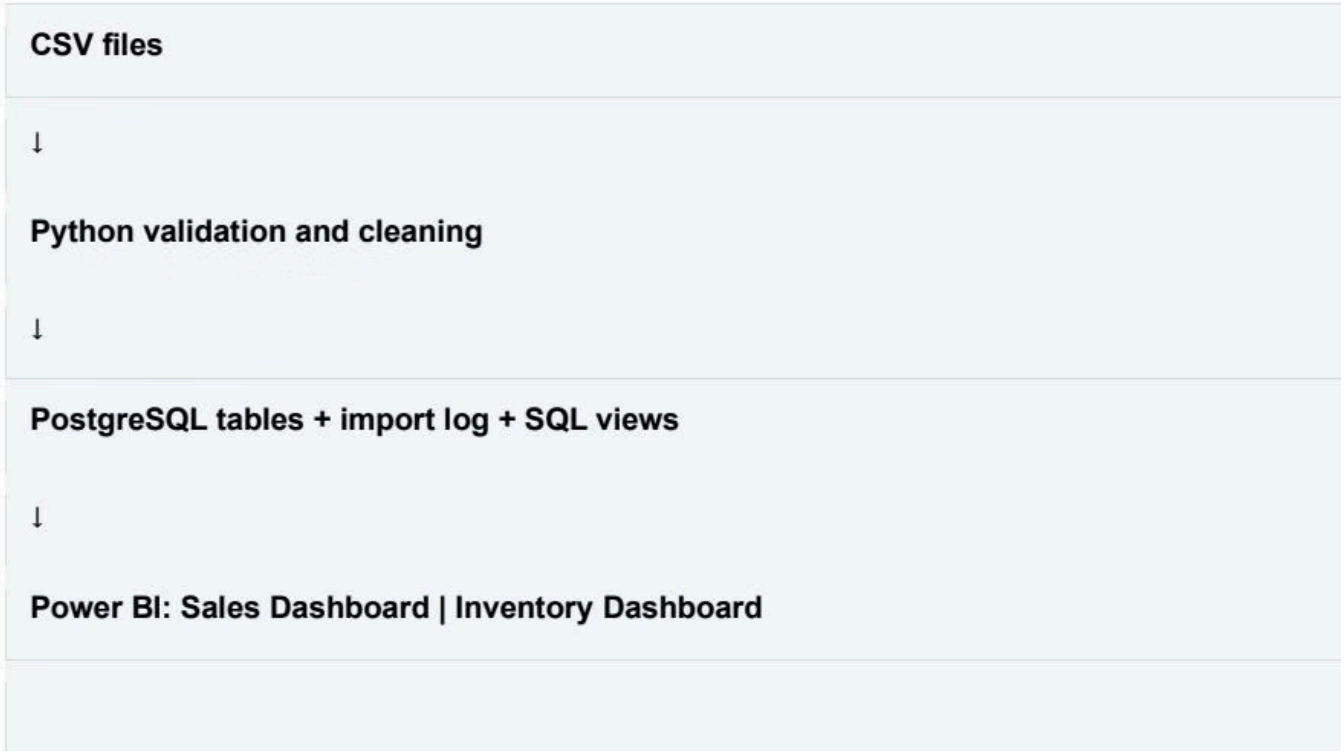
Files are placed in folders, checked with Python, and loaded into PostgreSQL.

Made with GAMMA

One pipeline connects files, systems, and decisions

Source files move through Python validation and cleaning, PostgreSQL storage and SQL views, then into Power BI sales and inventory reporting.

SOURCE → REPORTING



Supplied architecture flowchart.

Made with GAMMA

Python validates and cleans CSV exports

01

Validate

Unsupported file types, unreadable CSVs, empty files, missing columns, and missing required values.

02

Clean

Trim text, parse dates, convert numbers, and remove exact duplicate rows.

03

Load or reject

Check business rules and product references, append valid rows, and move files to processed or rejected folders.

04

Log

Log successful imports and use SHA-256 file hashes to help flag repeated files.

Made with GAMMA

PostgreSQL creates a durable analytical foundation

Layer	What it holds	Purpose
Core tables	Products, sales, inventory, and file_import_log.	Stores records and import history.
Analytical views	Sales joined to product details, plus inventory status and reorder logic.	Supports sales, product, and stock-status analysis.
Power BI model	Products acts as the shared dimension for sales and inventory analysis.	Links sales and inventory through product_id.

Made with GAMMA

Two planned Power BI pages answer key questions

Sales Performance

Sales trend, top 10 products, revenue by category, KPI cards, and payment methods.

Inventory Management

Stock availability, low-stock products, category stock, and reorder alerts.

Make the page easy to read

Use date, category, and payment slicers, aligned KPI cards, readable labels, and consistent number formatting.

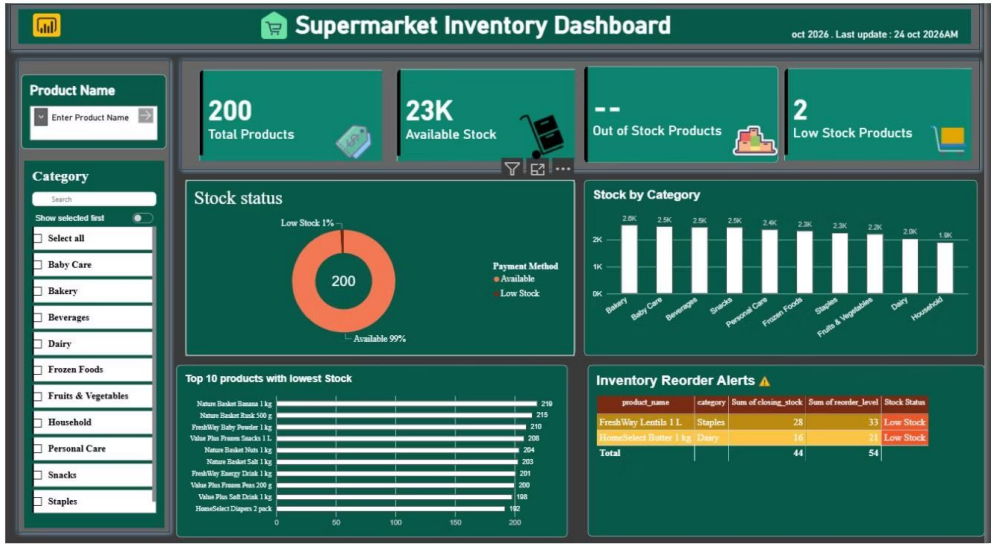
Made with GAMMA

The dashboards show which products need replenishment

The report includes illustrative sales and inventory dashboard snapshots to demonstrate the intended experience; KPI values and final visuals remain subject to end-to-end validation.



Illustrative sales dashboard snapshot.



Illustrative inventory dashboard snapshot.

Made with GAMMA

Trust comes from testing, not just dashboards

File behavior

Test valid, empty, unreadable, and structurally invalid CSV files.

Import behavior

Verify duplicate-file handling, unknown product references, failed inserts, and retry behavior.

KPI logic

Confirm discount units before publishing revenue or profit KPIs.

Refresh behavior

Confirm Power BI reloads from the selected source files; CSV-based reports do not reflect PostgreSQL changes automatically.

Made with GAMMA

A portfolio-ready foundation, with clear next steps

01 **Run an end-to-end test**

Use controlled sample files and test duplicate product IDs, failed inserts, and retries.

02 **Finish and capture the Power BI pages**

Complete both pages and capture screenshots after validation.

03 **Publish the foundation**

Reproducible code, sanitized samples, documentation, and environment-based credentials.

04 **Measure before claiming impact**

Track time saved and data-quality improvements before claiming production impact.

Made with GAMMA